



Smart wristband
be safe

Smart Wristband

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Abstract

Nowadays, Alzheimer's disease presents growing challenges in ensuring the safety and health of patients. Wandering outside designated safe zones and undetected health abnormalities, such as irregular heart rate or high body temperature, are critical issues faced by caregivers. Existing solutions often lack integration or fail to provide real-time alerts, highlighting the need for a device that can simultaneously monitor safety and health parameters. The study identifies a critical gap in the availability of integrated devices capable of providing both location tracking and health monitoring. Wandering outside safe zones can lead to accidents, while unmonitored health data like abnormal heart rate or temperature can result in delayed medical intervention. Existing devices are often fragmented or lack real-time alerts, creating an urgent need for a comprehensive solution.

The objective of this project is to design and develop a smart wristband specifically for Alzheimer's patients. The wristband combines location tracking, the ability to define safe zones, and real-time monitoring of vital signs such as heart rate and body temperature. It aims to alert caregivers through immediate notifications when a patient exits a safe zone or when abnormal health readings are detected. The wristband's operating system was developed using C++ and MATLAB, with mobile applications for Apple and Android ensuring seamless caregiver interaction. GPS technology enables accurate location tracking, and built-in sensors ensure reliable monitoring of vital signs. Notifications are transmitted via the app for timely intervention. The results demonstrate the wristband's ability to address critical safety and health concerns, enhancing patient safety, enabling prompt caregiver response, and improving the quality of life for both patients and caregivers.

في الوقت الحالي، يمثل مرض الزهايمر تحديات متزايدة في ضمان سلامة وصحة المرضى. يُعد التوهان خارج المناطق الآمنة المحددة واكتشاف الحالات الصحية غير الطبيعية مثل اضطراب معدل ضربات القلب أو ارتفاع درجة الحرارة من القضايا الحاسمة التي يواجهها مقدمو الرعاية. وغالباً ما تفتقر الحلول الحالية إلى التكامل أو لا تقدم تنبيهات في الوقت الفعلي، مما يبرز الحاجة إلى جهاز يمكنه مراقبة السلامة والمعايير الصحية في الوقت نفسه. تُحدد الدراسة فجوة حاسمة في توفر الأجهزة المتكاملة القادرة على تقديم تنبؤ الموقع ومراقبة الصحة. فقد يؤدي التوهان خارج المناطق الآمنة إلى الحوادث، بينما يمكن أن يؤدي عدم مراقبة البيانات الصحية مثل معدل ضربات القلب غير الطبيعي أو درجة الحرارة المرتفعة إلى تأخير التدخل الطبي. غالباً ما تكون الأجهزة الحالية مجزأة أو تفتقر إلى التنبيهات في الوقت الفعلي، مما يخلق حاجة ملحة لحل شامل. يتمثل هدف هذا المشروع في تصميم وتطوير سوار ذكي مخصص لمرضى الزهايمر. يجمع السوار بين تنبؤ الموقع، وإمكانية تحديد المناطق الآمنة، والمراقبة الفورية للعلامات الحيوية مثل معدل ضربات القلب ودرجة الحرارة. ويهدف السوار إلى تنبيه مقدمي الرعاية من خلال إشعارات فورية عند خروج المريض من المنطقة الآمنة أو عند اكتشاف قراءات صحية غير طبيعية. تم تطوير نظام تشغيل السوار باستخدام لغتي البرمجة C++ و MATLAB، مع تطوير تطبيقات موبايل لأنظمة Apple و Android لضمان تفاعل سلس مع مقدمي الرعاية. تنتج تكنولوجيا GPS تنبؤ الموقع بدقة، وتضمن المستشعرات المدمجة مراقبة موثوقة للعلامات الحيوية. يتم إرسال التنبيهات عبر التطبيق لضمان التدخل في الوقت المناسب. تظهر النتائج قدرة السوار على معالجة لقضايا الرئيسية المتعلقة بالسلامة والصحة، مما يعزز سلامة المرضى، ويسهم في استجابة سريعة من مقدمي الرعاية، ويحسن جودة الحياة لكل من المرضى ومقدمي الرعاية.

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1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that primarily affects elderly individuals, leading to cognitive decline, memory loss, and behavioral changes. According to the World Health Organization (WHO), dementia, with Alzheimer's disease as the most common form, affects millions of people worldwide, presenting significant challenges for both patients and caregivers. One of the most critical issues in managing Alzheimer's patients is ensuring their safety, particularly as the disease progresses and patients are prone to wandering, disorientation, and other health complications. In recent years, wearable health-monitoring devices have emerged as a promising solution to assist in managing the condition of Alzheimer's patients. These devices offer real-time tracking of vital health parameters and location, thereby improving patient safety and providing peace of mind to caregivers. Among these wearable technologies, smart wristbands have gained significant attention due to their portability, continuous monitoring capabilities, and ease of use. This project proposes the development of a smart wristband designed specifically for Alzheimer's patients, focusing on real-time health monitoring and safety features. The wristband integrates multiple sensors, including heart rate monitors, temperature sensors, and GPS, to provide continuous, real-time data on the patient's health and location. Furthermore, the wristband is paired with a mobile application that allows caregivers to monitor these parameters remotely. One of the key features of this system is the safe zone functionality, which enables caregivers to define a geographic boundary. If the patient moves outside this zone, an alert is sent, ensuring that timely interventions can be made. The primary objective of this project is to design and implement a cost-effective and user-friendly smart wristband that can serve as an early warning system for caregivers. It aims to enhance patient safety by detecting abnormalities in health indicators such as heart rate and temperature and preventing wandering by monitoring the patient's location within a safe zone. This system leverages the advancements in wearable technology, mobile applications, and cloud computing to create an integrated platform for real-time health monitoring. The project also contributes to the broader healthcare technology landscape by offering a scalable solution that can be extended to other healthcare conditions beyond Alzheimer's, where continuous monitoring is critical. By integrating multiple health-monitoring technologies into a single wearable device, this project aims to make a significant impact on patient care, caregiver support, and healthcare outcomes.

1.1 Problem Description and Motivation

Alzheimer's disease is considered one of the major challenges faced by many families, as it causes memory loss and cognitive decline, leading to significant difficulties in the lives of patients and their loved ones. One of the most prominent issues faced by patients is the risk of getting lost, as they may leave home without being aware of their location, or how to return. This situation creates considerable anxiety for their families, who constantly worry about their safety. Patients' need for security and support, along with families' desire to monitor them and ensure their safety, highlights the necessity for developing a smart wristband to track patients. Therefore, this study does try to overcome this issue.

1.2 Objectives

To design a Smart Wristband for Alzheimer's patients.

1.3 Digitization and Vision 2030

The project is in line with Vision 2030 by promoting innovation in the health sector and increasing the use of technology in patient care, which contributes to the promotion of public health and boosting the local economy.

The smart wristband project can have a significant impact on improving the care of Alzheimer's patients and enhancing safety and comfort for patients and their families, making it an innovative and relevant project for Vision 2030.

2. Literature Review

Several recent studies [1],[2],[3] have suggested that monitoring systems combining physiological and location-based data can significantly enhance health outcomes and safety for at-risk individuals.

In [1] did describe monitoring systems that can provide continuous physiological data, as well as better information regarding the general health of individuals. Thus, such vital-sign monitoring systems will reduce healthcare costs by disease prevention and enhance the quality of life. However, that study did not provide a specific smart wristband for Alzheimer's patients.

In [2], the authors investigated the development of Ambient Assistive Living (AAL) technologies to address the challenges posed by an aging population, emphasizing the need to reduce healthcare costs while improving the quality of care. The study introduced a smart healthcare monitoring system designed for the remote observation of elderly individuals, with a focus on tracking physiological data to detect specific disorders and enable early intervention practices. The proposed system demonstrated significant performance efficiency, achieving low latency (96% of packets received within less than 1 millisecond) and minimal packet loss (2.2% of total packets). However, while this study offers a valuable contribution to the field of remote health monitoring, it does not incorporate GPS tracking as a component of its system. GPS tracking could provide additional benefits, such as location-based alerts.

The authors in [3] proposed the development of a wearable smart locator band designed to monitor and track individuals. The device incorporates an AVR microcontroller, GPS module, GSM module, and a switching unit. This technology allows caregivers to track the wearer's location in real-time using an Android mobile application that displays the location on a Google Map. The device's capabilities have potential applications for individuals with conditions such as autism, dementia, or Alzheimer's disease, enhancing their safety and independence. However, this model lacks a defined safe zone feature, which could further enhance its functionality by alerting caregivers when the wearer exits specified areas.

3. Methodology

3.1 Requirements

3.1.1 Functional Requirements

The system must allow users to log in to their account.

The system must provide a user registration process if the user does not have an existing account.

The system must allow users to reset their password if they forget it.

The system must continuously track the patient's location using the wristband's GPS module.

The system must send an alert notification to the caregiver whenever the patient exceeds the boundaries of the predefined safe zone.

The system must monitor the patient's health data (heart rate and temperature) in real time.

The system must send a warning notification if the heart rate or temperature exceeds critical thresholds.

The system must allow the caregiver to define a safe zone through the mobile app.

The system must allow users to initiate a support chat within the mobile app.

3.1.2 Non-Functional Requirements

Performance: Real-time data monitoring with <2 seconds latency; app processes notifications in <5 second; wristband battery life of 48 hours.

Scalability: The backend supports up to 100,000 users without performance issues, handling 50% more users than initially planned, and the app accommodates multiple devices per caregiver.

Reliability: 99.9% system uptime; handles intermittent internet disconnections.

Security: End-to-end encryption, HIPAA compliance, and multi-factor authentication for caregivers.

Usability: Easy-to-understand notifications, setup time of <5 minutes.

Maintainability: Remote firmware updates with <2 minutes downtime; regular app updates.

Compatibility: Supports Android 8.0+, iOS 13+, Bluetooth 5.0+.

Environmental: The watch is designed to withstand extreme weather conditions, water-resistant (IP67).

3.2 Hardware

The hardware components of the smart wristband are designed to support continuous health monitoring and provide real-time data transmission to the mobile app. These components ensure the wristband can send notifications for abnormal events such as the patient leaving the safe zone, heart rate irregularities, or elevated body temperature.

1- Microcontroller: The ESP32 microcontroller manages communication between the wristband's sensors and the mobile application. It is responsible for collecting GPS, heart rate, and temperature data and transmitting them to the mobile app via Bluetooth Low Energy (BLE). The ESP32 also processes data locally to identify if any thresholds have been breached, such as exceeding safe zone boundaries or detecting abnormal heart rate or temperature values.

2- Sensors

2.1 GPS (GPS Sensor): (e.g., u-blox NEO-6M GPS Module)

To track the patient's geographical location in real-time and sends the location coordinates to the application.

2.2 Heart Rate Sensor: (e.g., MAX30100 Pulse Oximeter and Heart Rate Sensor)

Measures the patient's heart rate per minute and helps monitor heart health and detect any abnormal changes in pulse.

2.3 Temperature Sensor: (e.g., DHT22 (AM2302) Temperature and Humidity Sensor)

Measures body temperature and helps detect fever or other health issues related to abnormal temperature.

3- Long-lasting Battery:

Ensures the wristband operates for extended periods without frequent recharging.

4- Comfortable and Waterproof Design:

Safe and comfortable materials for daily use, with waterproofing properties.

5- Cellular Connectivity: (e.g., GSM/4G LTE)

It ensures the ability to send alerts and data remotely, even when there is no nearby phone connection.

6- Small OLED Display and Buttons:

Facilitates ease of use with a small screen and simple buttons.

7- Charging Base:

Supports easy charging of the wristband (either wired or wireless).

3.3 System & Software

1- Mobile Application

An application for Android and iOS using technologies like Flutter or React Native.

Displays real-time location and monitors vital signs.

Features an easy-to-use interface with the ability to set up safe zone settings.

2- Database

Stores user data, location, and vital signs in a database such as MySQL.

SQL queries are used to retrieve and organize the data into separate tables.

3- Server

Runs on Ubuntu Server with a quad-core processor and 16GB of RAM.

Processes data from the smart wristband using Node.js or Django and sends it to the database and application.

4- Security

Data is encrypted during transmission using TLS/SSL and stored encrypted in the database.

Uses OAuth or JWT for authentication and protects the server from attacks with Firewall and security techniques such as Security Headers.

5- Embedded Firmware

The ESP32 microcontroller, programmed in C++, handles data acquisition and processing.

6-Mobile App Backend and API

The RESTful API ensures real-time synchronization between the app and the server, managing data flow and notifications.

3.4 User Level

Admin level:

The admin can create an account, but if he has an account, he can log in.

The admin can respond to users inquiries.

User level:

The user can create an account, but if he has an account, he can log in.

The user can edit the profile. (e.g. PhoneNum, Username, Age, save zone)

The user can define save zone and update it.

The user can measure statistic vital signs (e.g. Heart Rate, temperature)

The user can receive warning notifications (e.g. The patient leaves the save zone, abnormal heart rate low temperature).

The user can track the patient's location on the map.

3.5 Data and Contents

Personal Data: Patient's identification details, such as name, age.

Health Data: Readings from medical devices like heart rate, body temperature.

Geographical Data: Location information using GPS to track and pinpoint the patient's location.

Alerts and Signals: Notifications for the patient and caregivers in case of any abnormal changes in movement or location.

3.6 Project Management Strategies Involved in the Project

Feasibility Study and Comprehensive Planning

- **Define Objectives and Requirements:** conducting a detailed study to outline the essential features for Alzheimer's patients and caregivers, such as GPS, emergency alerts, and health monitoring.
- **Financial and Resource Planning:** Establish an accurate budget covering development, production, and marketing costs, and ensure necessary funding.

Scope Management

- Clearly defining the project scope is crucial to maintaining focus on core objectives, such as the wristband essential features and a user-friendly interface. Implementing a system to control scope changes to avoid unnecessary additions that may increase costs or time.

3.7 Project Scope

This project focuses on developing a smart wristband designed for Alzheimer's patients, featuring comprehensive functions for location tracking and vital sign monitoring. The wristband will use Global Positioning System (GPS) technology to track patients' locations in real-time, ensuring their safety at home. Additionally, it will monitor vital signs such as heart rate and blood pressure, providing crucial health data for caregivers. The device aims to assist patients in navigating their surroundings while providing peace of mind for their families and caregiver

3.8 Role of each project team member

UML Diagrams:

Use-Case: Afnan, Raghad, Rahaf, Razan, Sara, Hanan

Use-Case Description: Afnan, Raghad

DFD: Hanan and Sara

ER: Rahaf, Razan and Afnan

Sequence: Afnan, Raghad

Activity: Sara, Raghad, Afnan

User Interface Design: Razan

Smart wristband Prototype: Hanan

Documentation and Research: In terms of research and studies, Razan was particularly involved in collecting them.

All team members contributed to documenting and discussing all the contents, as the team worked together step by step, reviewing all aspects of the project.

3.9 Time Plan of the Project

Task Name	SEPTEMBER				OCTOBER				NOVEMBER				December			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
Choose the topic of project																
Search about the topic																
Abstract																
Introduction																
Problem Description and Motivation																
Objectives																
Digitization and Vision 2030																
Literature Review																
Requirements																
Hardware																
System & Software																
User Level																
Admin Level																
Data and Contents																
Project Management Strategies Involved in the Project																
Design Overview (UI/UX)																
Diagrams																
Results and Discussion																
Conclusions and Future Work																

Table 1: Time line of the project

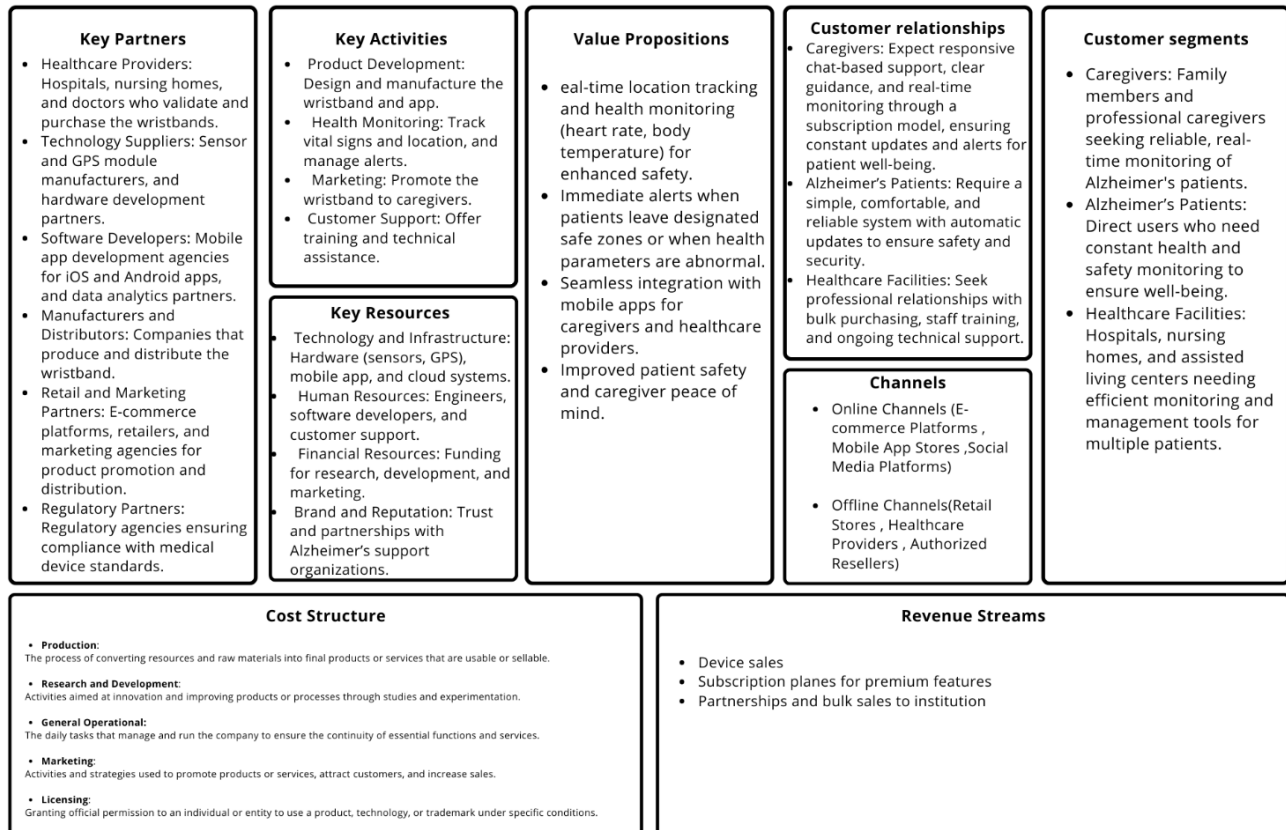
3.10 The Gantt chart describes

A horizontal bar chart used in project management to visually represent a project plan over time.



4. Suggested Business Modal

BUSINESS MODEL



4.1 What are the Business Benefits / Proposals of the Project?

1. Enhancing Independence: The smart wristband helps individuals with Alzheimer's maintain their independence, boosting their confidence and comfort.
2. Reducing Care Costs: By decreasing the need for constant monitoring by caregivers, the technology can help lower associated care costs.
3. Improving Personal Safety: Features like location tracking and alerts ensure the safety of patients and reduce accident risks.
4. Providing Sustainable Support: Establishing ongoing support services strengthens the relationship between the brand and customers, fostering loyalty and satisfaction.
5. New Funding Opportunities: With a focus on health technology, the project can attract investments from healthcare investors.

5. Materials and Methods

a. Operating System

In the Alzheimer's smart wristband project, an Embedded Real-Time Operating System (RTOS) is utilized to efficiently manage the complexities of sensor data collection, processing, and communication within the smart wristband environment. The operating system for the wristband application will depend on the target platform. For mobile app development, iOS or Android will be used to complement the wristband's functionality and provide a seamless user experience.

b. Documentation

All project processes and designs are documented using Google Docs, including hardware schematics, software development logs, and testing protocols. The documentation provides a clear and traceable record for development and future enhancements.

c. UML Designing

UML (Unified Modeling Language) is a standardized modeling language used to represent the design and behavior of a system. For the Alzheimer's smart wristband, the relevant UML diagrams include the Use Case Diagram, Activity Diagram, Sequence Diagram which describe the system's structure, interactions, and functionality.

d. Graphics and Designs

The graphic design project for a smart wristband targeting Alzheimer's patients focuses on creating an easy and lightweight device for people with Alzheimer's and their caregivers. The purpose includes monitoring health through vital signs, providing location tracking via GPS, and the ability to send warning notifications to the application. The design will include a simple and intuitive user interface with soft and calm colors, clear and large fonts for easy reading and navigation, and illustrations of the benefits of the wristband. The user will be given the ability to reach the patient, monitor vital signs, and track their location through a map. The application will include a list of warning notifications as a table for the previous days and the current day.

e. Development IDE

This project uses various IDEs: MATLAB for algorithm development and system simulation, SIMULINK for modeling, and STATEFLOW for state-based control. The Arduino IDE programs the wristband's microcontroller, while Xcode and Android Studio develop the iOS and Android apps, respectively. These tools ensure efficient integration of hardware and software across platforms.

f. Language of Programming

The project uses C++ for the wristband's operating system, enabling efficient hardware interaction and real-time data processing. Swift powers the iOS app for smooth user experience and integration with Apple devices, while Java/Kotlin is used for the Android app, ensuring real-time data sync and a responsive interface. MATLAB aids in data analysis and simulation to test and validate the system's performance. These languages ensure a fully integrated, efficient solution.

g. Database

The project employs MySQL as the database management system to store and retrieve patient health data. MySQL is chosen for its robust performance, scalability, and ability to handle structured data efficiently. It is integrated with the backend server to ensure secure and real-time data access.

h. Web-Based Server / Other Engines

The project relies on a web-based server as the backbone of communication between the application, the wristband, and the database. This server is built using technologies such as Node.js or Django, providing a robust and scalable backend environment. The server manages real-time data transmission, ensuring efficient retrieval and updating of patient information and location data. It also supports secure communication protocols like HTTPS to protect sensitive data during transmission.

i. Hosting of the Project

The project utilizes reliable cloud hosting services, such as Google Cloud, to ensure continuous operation and support real-time data processing. The hosting platform offers features like automated backups, load balancing, and distributed servers to ensure high availability and scalability. Security is reinforced through HTTPS protocols and advanced firewalls to protect data from potential cyber threats.

5.1 Design Overview (UI/ UX)



Figure 1

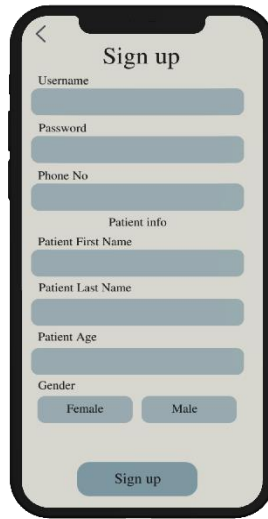


Figure 2



Figure 3

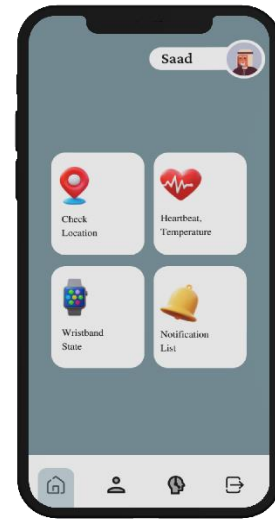


Figure 4

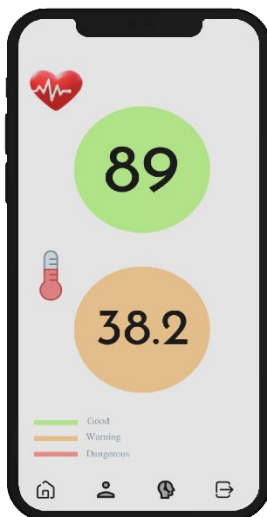


Figure 5

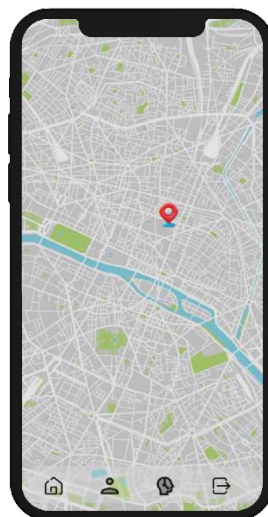


Figure 6

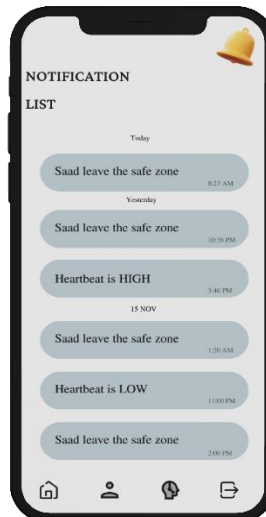


Figure 7

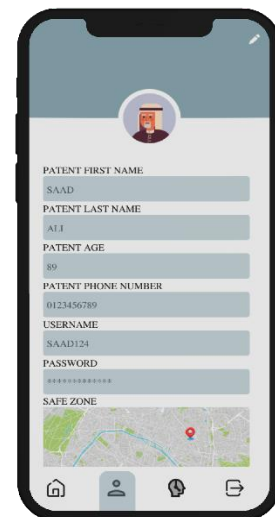


Figure 8



Figure 9



Figure 10

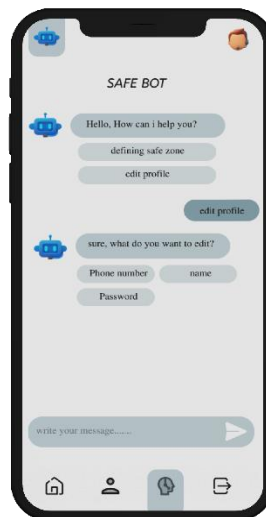


Figure 11

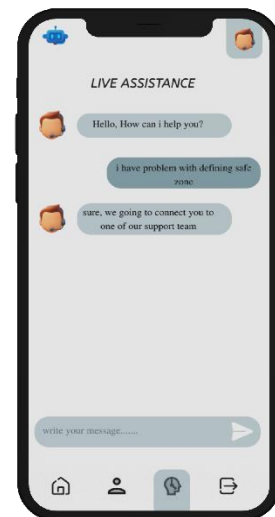


Figure 12

5.1.1 Developer Perspective

From a developer's perspective, the design of the smart wristband application should be modular and maintainable. Each component, such as user interface, health data tracking, and location services, should be developed as independent modules. This modularity allows for easy updates and enhancements without disrupting the overall system. Additionally, the application must be scalable to handle multiple users simultaneously, ensuring that real-time alerts and data synchronization remain efficient.

5.1.2 User Perspective

From the user's point of view, the design of the smart wristband app should prioritize simplicity and accessibility. The interface should be intuitive, allowing their caregivers to navigate easily. Key features, such as location tracking and emergency alerts, should be featured prominently and easily accessible. The app should also provide updates, enhancing the user experience and enhancing a sense of security for both caregivers.

5.2 Diagrams

5.2.1 Use Case Diagram

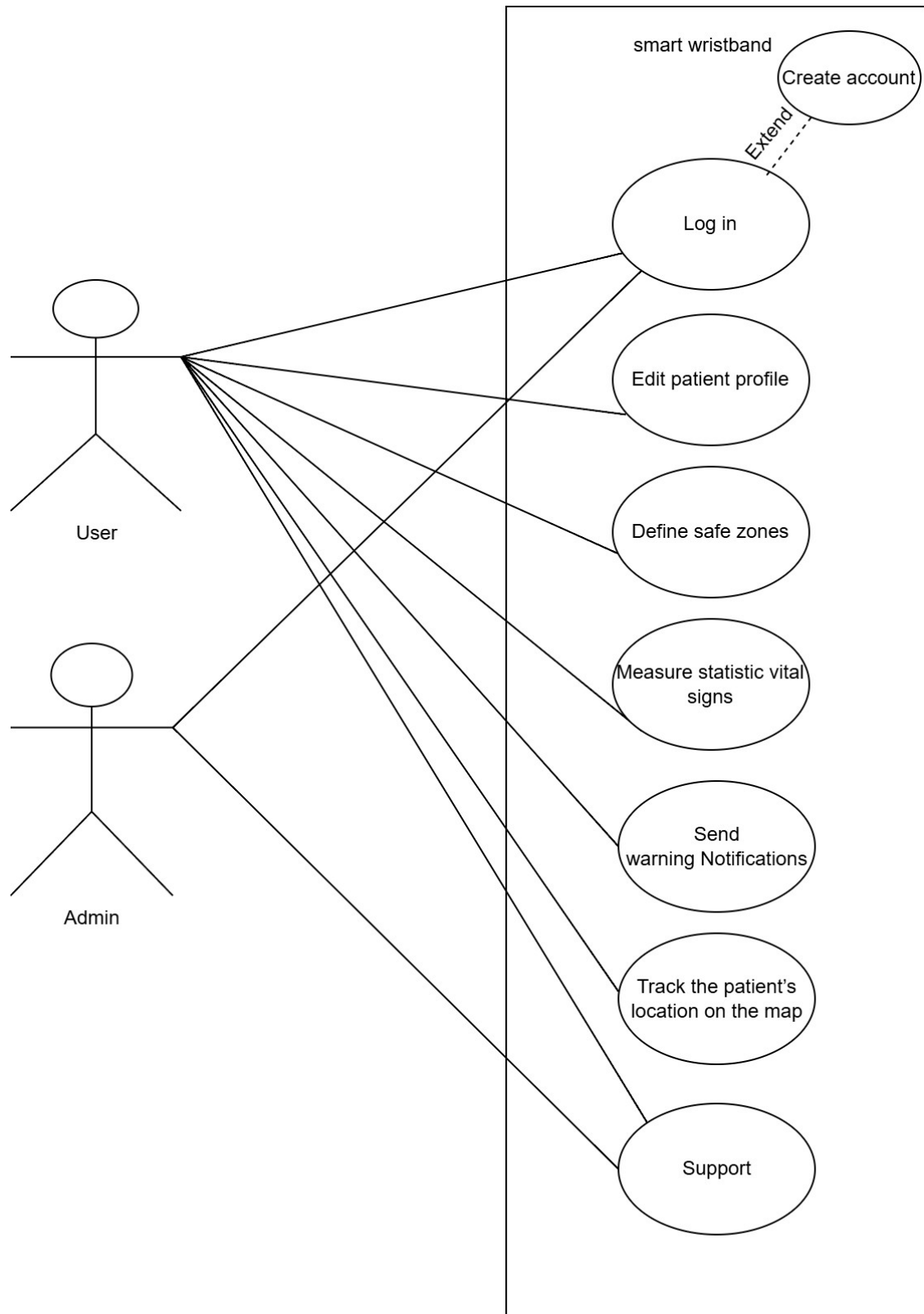


Figure 13

5.2.2 Use Case Description

Use case name: Track the Patient's Location on the Map	Id: 7	Important level: High
Primary actor: User	Use case type: Essential, Functional	
Stakeholders and Interests: User: Needs to know the real-time location of the patient, especially in emergencies or when the patient exits a predefined safe zone. Interested in having accurate and easily accessible information through the app. Alzheimer's Patient: Ensuring their safety by allowing family members to locate them if they get lost. Admin: Ensures that the tracking feature functions accurately and reliably to meet user expectations.		
Brief Description: This use case allows the user to monitor the patient’s current location on a map via the mobile app connected to the smart wristband. In case of an emergency or when the patient exits a predefined safe zone, the system displays the patient’s live location on the map within the app.		
Trigger: The patient's exit from the predefined safe zone.		
Type: Monitoring & Query		
Relationship: Include: Display the location on a map using GPS. Extend: Send an alert if the patient leaves the safe zone.		
Normal flow of event: 1- The user logs into the application using their registered information. 2- The patient exits the predefined safe zone. 3- An immediate warning notification is sent to the user. 4- The application connects to the wristband to retrieve real-time location data using the GPS system. 5- The user tracks the patient's life location through the application.		
Sub-flow: 1- If the wristband's battery is low, a notification will be displayed to the user. 2- In case of GPS signal loss, a notification will be sent to the user, and the application will attempt to reconnect.		

5.2.3 Data Flow Diagram

Context Level

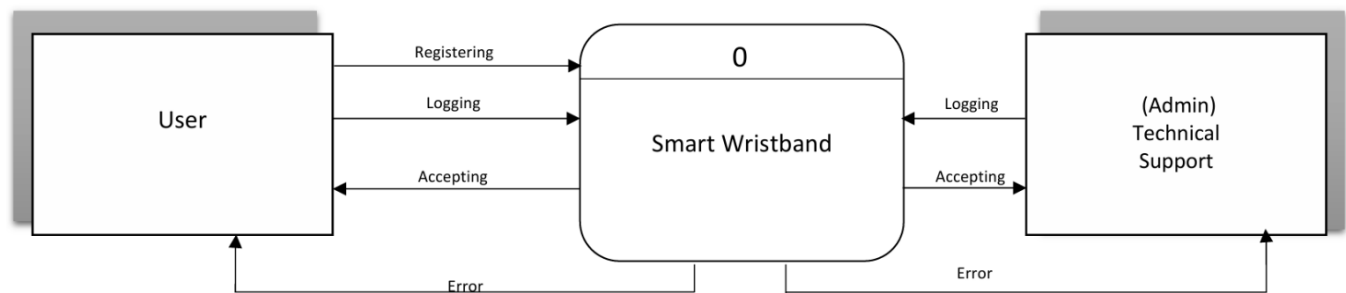


Diagram 0

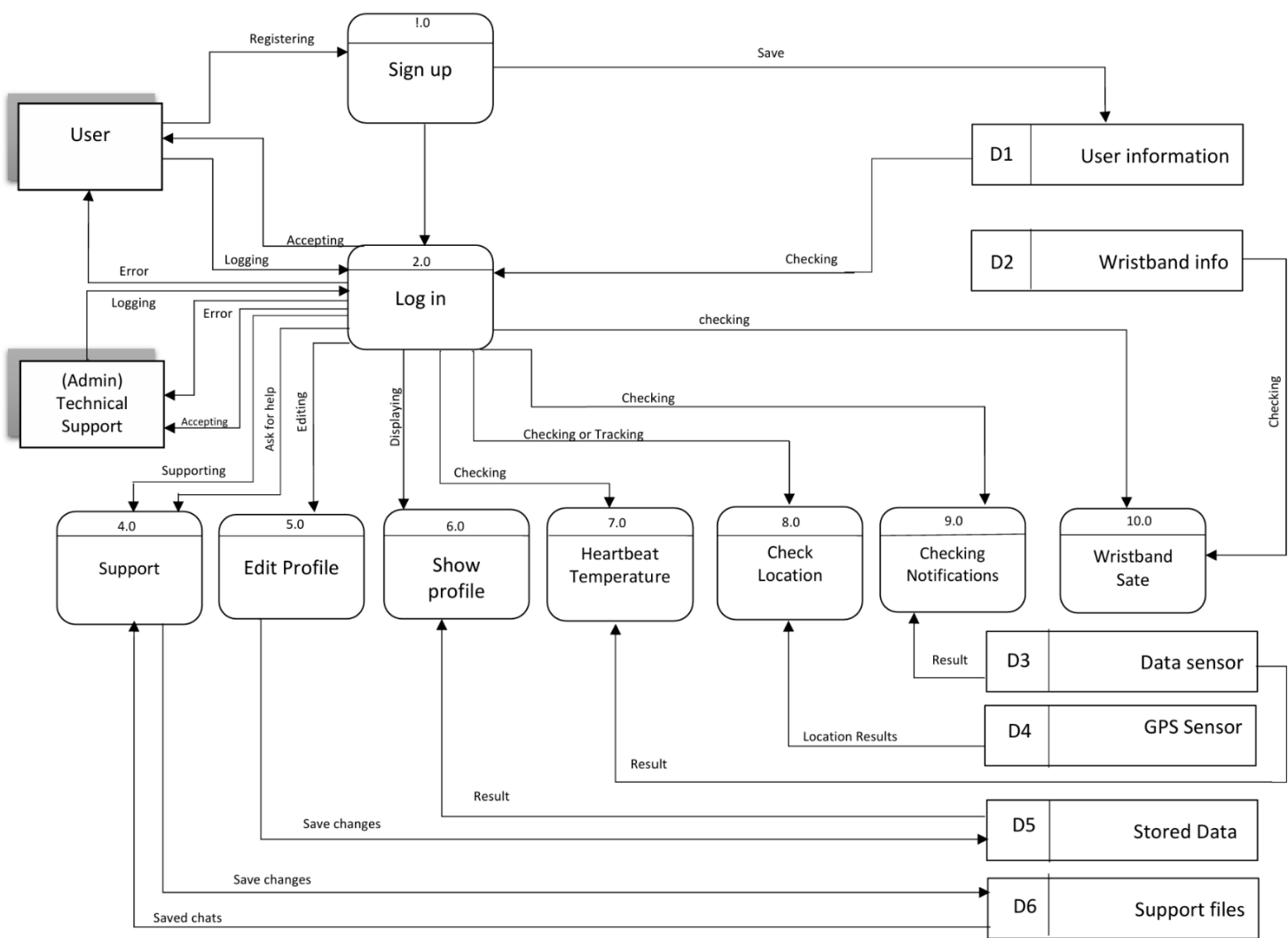


Figure 15

5.2.4 Sequence Diagram

Track the patient's location on the map

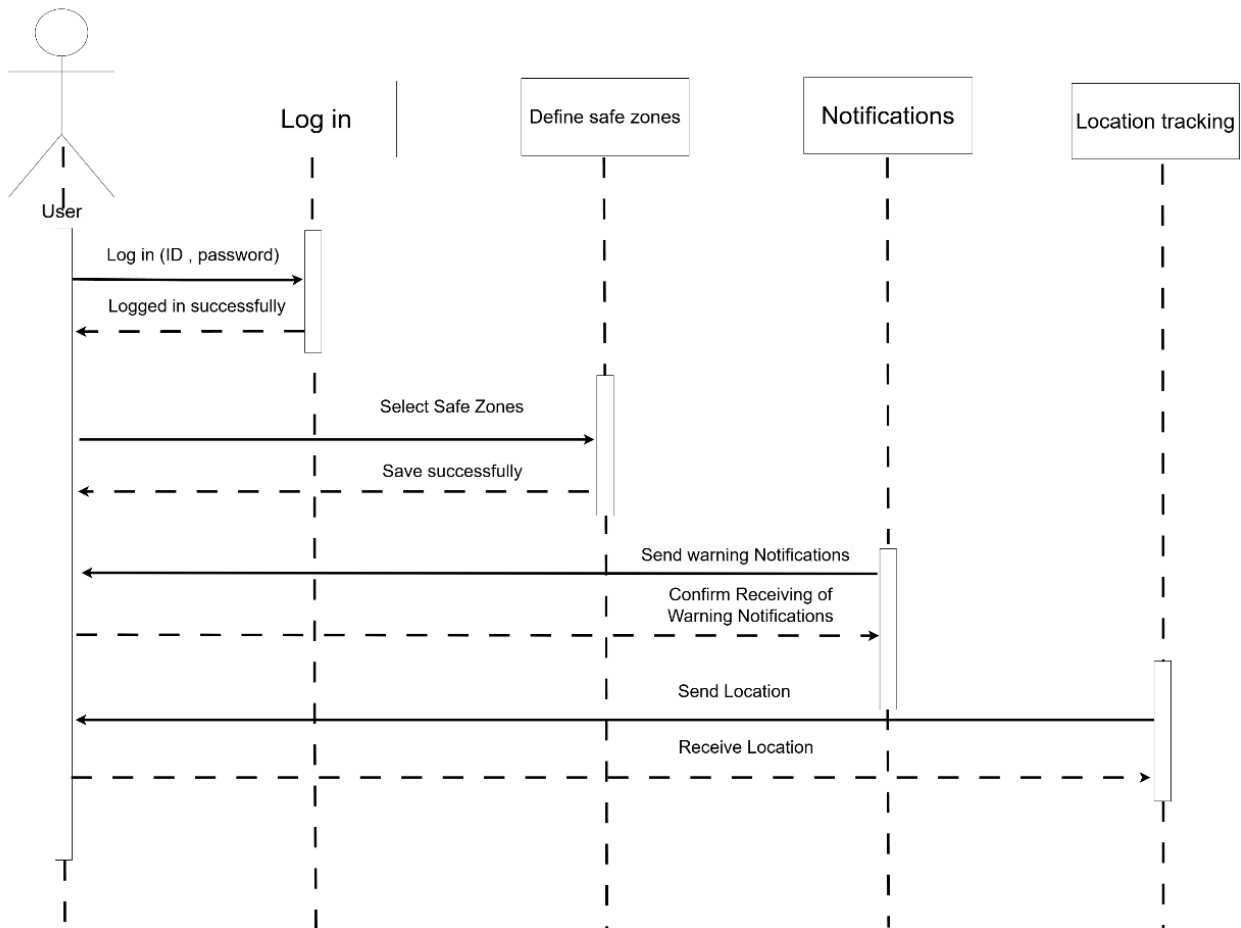


Figure 16

5.2.5 Activity Diagram

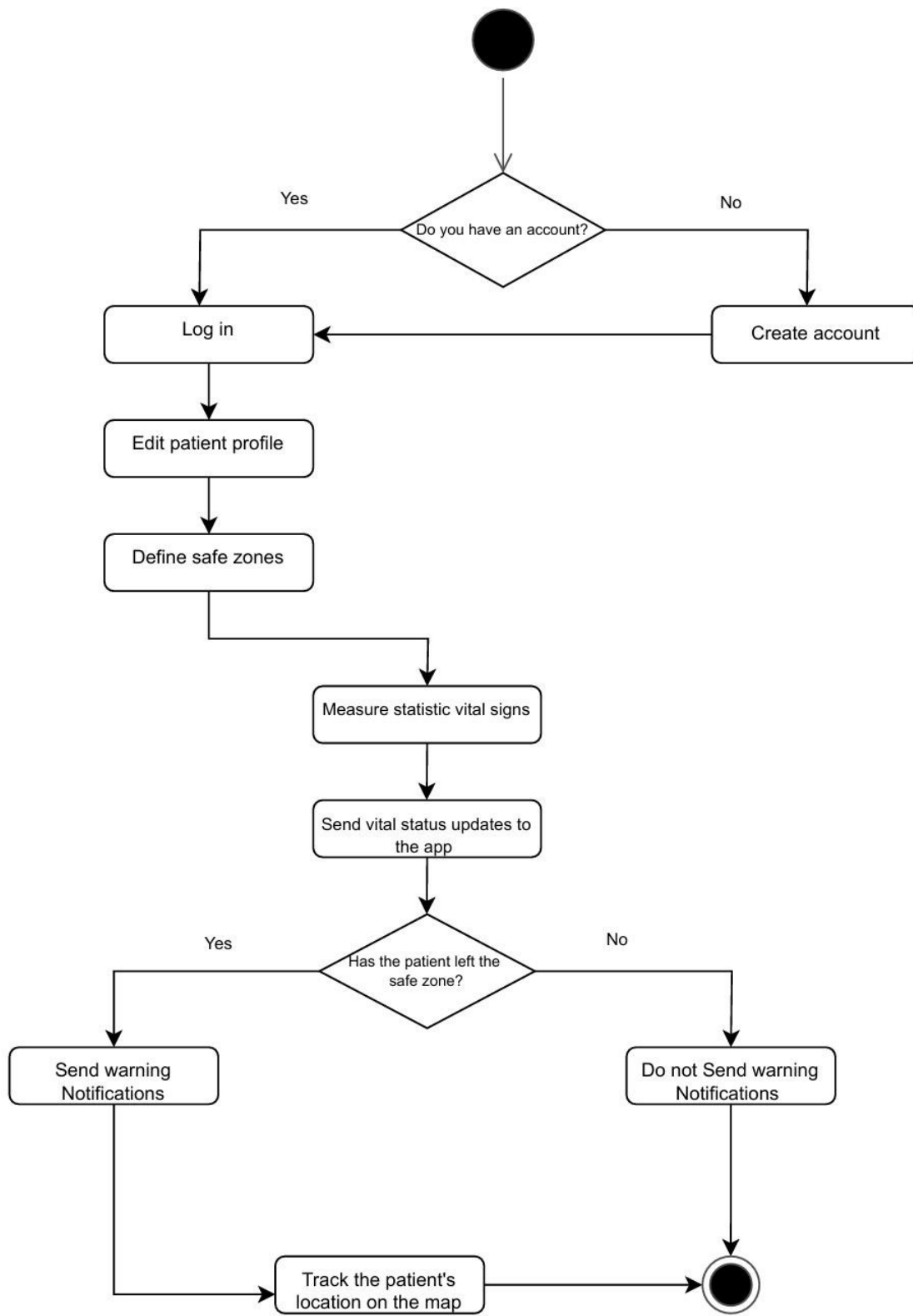


Figure 17

5.2.6 ER Diagram

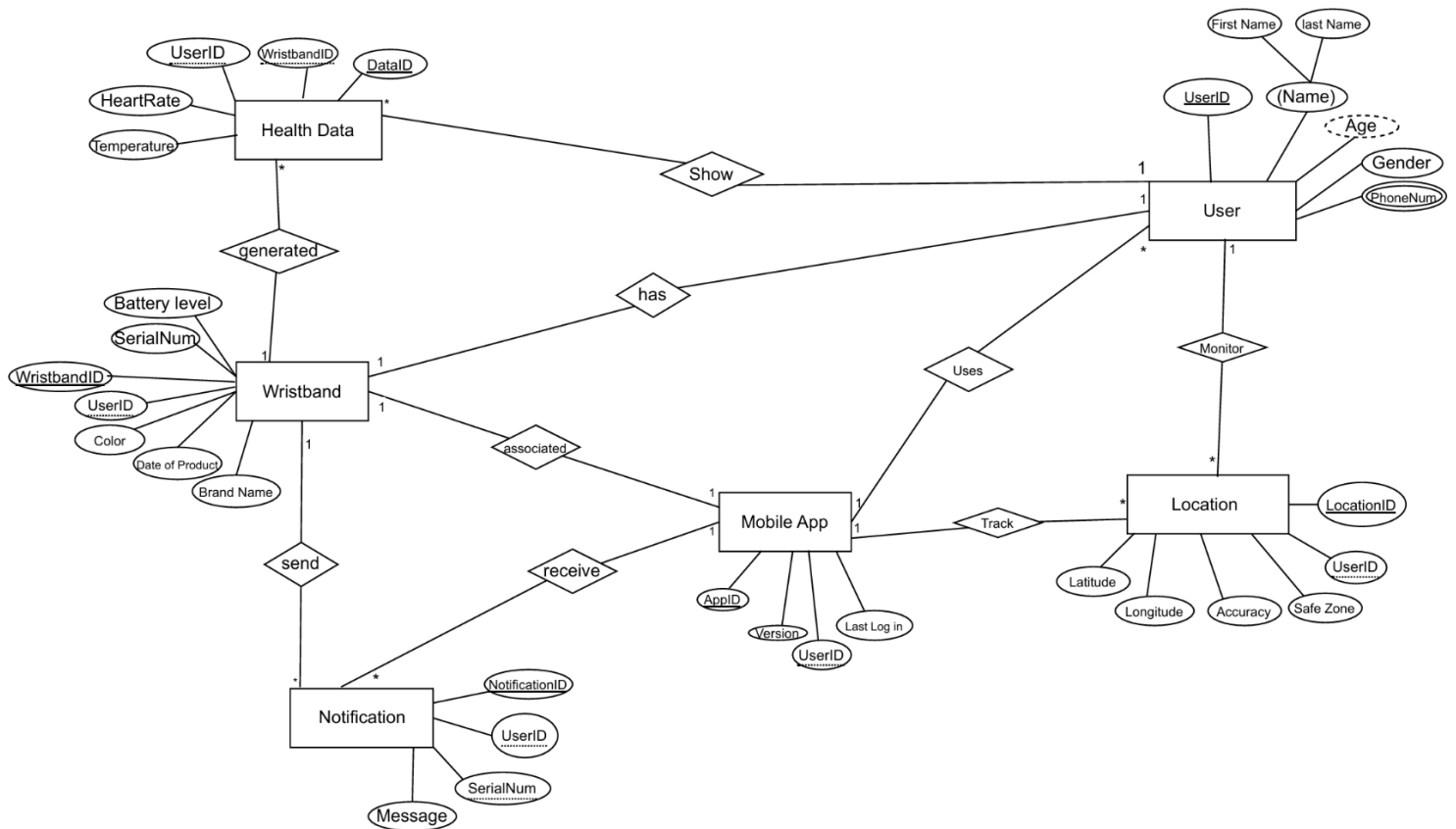


Figure 18

6. Results and Discussion

The project results include various interfaces designed to provide caregivers with a user-friendly and comprehensive application for managing Alzheimer's patients. The Login and Signup Interfaces (Figures 1, 2, and 3) serve as the application's secure entry point, enabling caregivers to log in or create an account. These interfaces also allow users to register patient details, such as name, age, gender, and phone number. The Home Screen Interface (Figure 4) acts as the central navigation hub, offering seamless access to key features like health monitoring, GPS tracking, and notifications. The Health Monitoring Interface (Figure 5) displays real-time data, including heart rate and body temperature, and alerts caregivers if these metrics exceed safe thresholds. The GPS Tracking Interface (Figure 6) provides live location updates, ensuring caregivers are notified immediately if a patient exits the predefined safe zone. The Notification List Interface (Figure 7) organizes alerts in chronological order, complete with timestamps for event tracking. The Edit Profile Interface (Figure 8) allows caregivers to manage personal and patient details, including safe zone configurations. To ensure the wristband remains functional, the Wristband Status Interface (Figure 9) monitors the battery status and displays the device's unique ID. For quick and efficient setup, the QR Code Scanning Interface (Figure 10) streamlines pairing the wristband with the app. Finally, the Chatbot and Live Support Interfaces (Figures 11 and 12) offer a dual-layer support system, where the chatbot handles common queries, and the live support feature provides real-time assistance for more complex issues. Together, these interfaces create a robust and reliable system that enhances patient care and monitoring, ensuring safety, health tracking, and caregiver support in real time.



Figure 19



Figure 20

To evaluate the effectiveness of the proposed smart wristband system, a comprehensive simulation study was conducted, as shown in Figure 19. The simulation environment replicated real-world scenarios, incorporating factors such as GPS accuracy and varying user activity patterns. The results demonstrated that the wristband system performed with a high level of accuracy in detecting zone exits, achieving an average alert delivery time of 5 seconds under ideal conditions. Furthermore, the system showed robust performance across a range of user activities and environmental conditions, confirming its reliability for practical use.

Figure 20 show the design of the 3D Smart Wristband for Alzheimer's patients includes: a comfortable, adjustable silicone strap, integrated sensors to measure heart rate and temperature and GPS tracking, water-resistant and lightweight design, operates without a screen for ease of use.

7. Conclusions and Future Work

This project introduces a smart wristband system designed to enhance the care and safety of Alzheimer's patients. By integrating real-time health monitoring (including heart rate and temperature sensors) and GPS tracking, the system offers a reliable solution for caregivers to continuously track patient health and location. Additionally, the implementation of safe zone monitoring and automatic alert notifications ensures timely intervention in case of emergencies, thus significantly improving patient safety.

7.1 Future Work

While the current system demonstrates its effectiveness in monitoring and alerting caregivers, future work will focus on expanding the app's capabilities to manage multiple patients, incorporating motion sensors for fall detection, and enhancing interactive features for greater patient engagement. These improvements aim to increase the system's usability and effectiveness, making it a more comprehensive tool for both patients and caregivers. Additionally, a future enhancement will be the integration of an Emergency Assistance Feature, which directly connects the wristband to emergency services (such as 911 or local equivalents) in case of severe health issues or when the caregiver is unable to respond. This feature will enable the system to share the patient's location and health status directly with emergency teams, ensuring timely intervention. Another anticipated enhancement is the Audio Communication Feature, which will allow the wristband to include a microphone and speaker for voice communication between the patient and the caregiver. This capability will be especially useful if the patient becomes lost or confused, enabling effective guidance and reassurance to be provided in such situations.

7.2 Conclusions

In conclusion, this project lays the foundation for a comprehensive wearable health solution, contributing to the growing field of telemedicine and wearable health technologies. By improving real-time monitoring and caregiver support, the system enhances the quality of life for Alzheimer's patients and supports caregivers in managing complex care needs. The future direction of the project will further refine these features, ensuring that the system can meet the evolving demands of Alzheimer's care in the coming years.

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